ARRAY BASIC CONCEPTS

PREPARED BY:

KARLO EMIL MORALES

ARRAY

Is a data structure which allows you to store a collection of data, given that they are of the same data type. Array can store multiple elements by assigning a position for each element. This is also called the index number or position of the array.

INDEX -> Refers to the position or assignment reference of an element in the array.

50	22	31	54	33
Index 0	Index 1	Index 2	Index 3	Index 4

INDEX -> Refers to the position or assignment reference of an element in the array. The first element is always assigned to index 0 and the last element refers to the size or #of elements-1 in the array.

Always remember that the references of the elements is always equivalent to their assigned index number.

CREATING A SINGLE DIMENSIONARRAY

An array can be declared using different approaches:

int mynums[6];

The most important character here is the bracket, this indicates that what you are trying to declare is an array of integers.The name of the array is mynums and based on the declaration provided it is an array which can hold up to 6 elements.6 is the maximum capacity of the array created on the example.This array by default has no initial value.

int mynums[6] = {10,22,44,2,56,14};

Similar with the approach above this array has a different syntax.The 2nd declaration is still an array.However, the declared array already has a constant value which is indicated inside the { } symbol and separated using commas(,).This array has a size of 6 since it has 6 elements inside.

ACCESSING ELEMENTS FROM THEARRAY

Assume we have an array set of:

50	22	31	54	33
Index 0	Index 1	Index 2	Index 3	Index 4

```
int mynums[5]={50,22,31,54,33};
```

In order to access or use the elements stored in an array it is important to take note of the index number.Assume we want to display the number 54 from the array.

```
printf(“%d”,mynums[3]);
```

Since the target element in the array is 54 and it is located on index 3.The element can easily be accessed by calling the name of the array(mynums) and indicating what is the index number of the element which on our case is 3.

ASSIGNING VALUES TO AN ARRAY

Assume we have an array but with no initial value. How can we assign values inside the array?

```
int mynums[4];
```

The declared array currently has no values. But was declared to be able to store the maximum of 4 int values. The array can be assigned values by indicating the index number to store the value. Since we have 4 as our max size.We can assume that the indexes 0,1,2,3 are usable and we can assign values to those indexes.

```
mynums[2]=58;
```

It is very simple to assign values to an array. Indicate the name of the array and then indicate using bracket the index number for the assigned value.This code shows that 58 will be assigned to index 2 of array mynums.

RULES IN IMPLEMENTING ARRAYS

- *An array can be declared as any data type.*
- *An array must have a static or maximum size indicated upon declaration.*
- *An array must contain the same data type as how it was declared. You cannot use multiple data types in one declaration.*
- *You cannot access values which does not belong within an array.*
- *You cannot access index numbers which are above the maximum size declared. Eg. Array size is 8 you cannot access index numbers higher than 7.*
- *You can only assign a value identical to the data type of the array*
- *Index count always start with 0.*

TRY IT YOURSELVES

Create an array using all the data types.

Create a mini program which will allow the users to enter the values through the use of scanners the possible values of an array.

Create an array which has no values and try to access and index with no value.

THANK YOU.